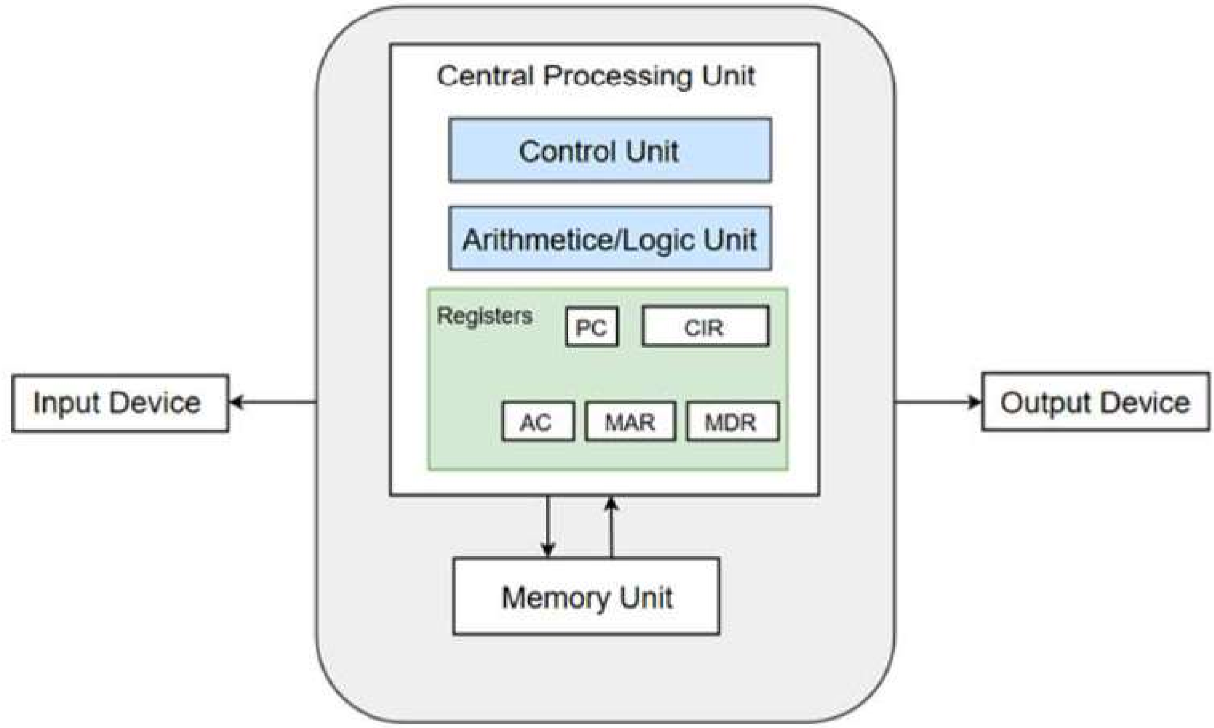
# M1.01 FUNCTIONAL UNITS




- **Central Processing Unit**
  - Control Unit
  - ALU
  - Registers
- **Input and Output Unit**
  - **Memory Unit**
  - RAM
  - ROM
- **Buses**
  - Address Bus
  - Data Bus
  - Control Bus

## Central Processing Unit

- The part of the Computer that performs the bulk of data processing operations is called the Central Processing Unit and is referred to as the CPU.
- The major components of CPU are Arithmetic and Logic Unit

(ALU), Control Unit (CU) and a variety of registers.

**Arithmetic and Logic Unit (ALU)**

▪ The Arithmetic and Logic Unit (ALU) performs the required microoperations for executing the instructions.

▪ In simple words, ALU allows arithmetic (add, subtract, etc.) and logic (AND, OR, NOT, etc.) operations to be carried ouT.

**Control Unit**

▪ The Control Unit of a computer system controls the operations of components like ALU, memory and input/output devices.

▪ **Registers**

▪Registers refer to high-speed storage areas in the CPU. The data processed by the CPU are fetched from the registers.

**Memory Unit**

▪A memory unit is a collection of storage cells together with associated circuits needed to transfer information in and out of the storage.

▪The memory stores binary information in groups of bits called words.

▪Two major types of memories are used in computer systems:

- ▪**RAM (Random Access Memory)**
- ▪**ROM (Read-Only Memory)**

**Buses**

▪ Buses are the means by which information is shared between the registers in a multiple-register configuration system.

▪A bus structure consists of a set of common lines, one for each bit of a register, through which binary information is transferred one at a time.

▪ Control signals determine which register is selected by the bus during each particular register transfer.

▪Von-Neumann Architecture comprised of three major bus systems for data transfer.

## M1.02 Basic Operational Concepts In Computer Organization

A Computer has five functional independant units like Input Unit, Memory Unit, Arithmetic & Logic Unit, Output Unit, Control Unit.

**Input Unit :-**

Computers take coded information via input unit. The most famous input device is keyboard. Whenever we press any key it is automatically being translated to corresponding binary code & transmitted over a cable to memory or proccessor.

**Memory Unit :-**

It stores programs as well as data and there are two types- Primary and Secondary Memory

Primary Memory is quite fast which works at electronic speed. Programs should be stored in memory before getting executed. Random Access Memory are those memory in which location can be accessed in a shorter period of time after specifying the address. Primary memory is essential but expensive so we went for secondary memory which is quite cheaper. It is used when large amount of data & programs are needed to store, particularly the information that we dont access very frequently. Ex- Magnetic Disks, Tapes

**Arithmetic & Logic Unit :-**

All the arithmetic & Logical operations are performed by ALU and this operation are initiated once the operands are brought into the processor.

**Output Unit** :– It displays the processed result to outside world.

**<u>Basic Operational Concepts</u>**

- Instructions take a vital role for the proper working of the computer.
- An appropriate program consisting of a list of instructions is stored in the memory so that the tasks can be started.
- The memory brings the Individual instructions into the processor, which executes the specified operations.
- Data which is to be used as operands are moreover also stored in the memory.

***Example:***

***Add LOCA, R0***

- ***This instruction adds the operand at memory location LOCA to the operand which will be present in the Register R0.***

- ***The above mentioned example can be written as follows:***

***Load LOCA, R1***

***Add R1, R0***

- ***First instruction sends the contents of the memory location LOCA into processor Register R0, and meanwhile the second instruction adds the contents of Register R1 and R0 and places the output in the Register R1.***

***The memory and the processor are are swapped and are started by sending the address of the memory location to be accessed to the memory unit and issuing the appropriate control signals.***

- ***The data is then transferred to or from the memory.***

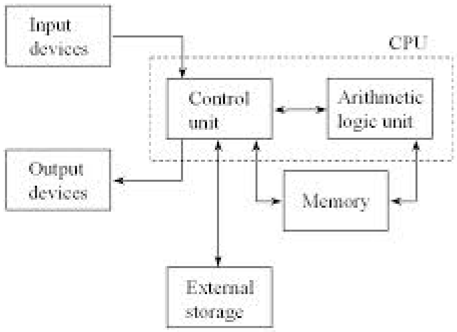
## M1.04 Explain the connection of memory to the processor.

**Analysing how processor and memory are connected :**–

- Processors have various registers to perform various functions :-
- Program Counter :- It contains the memory address of next instruction to be fetched.
- Instruction Register:- It holds the instruction which is currently being executed.
- MDR :- It facilities communication with memory. It contains the data to be written into or read out of the addressed location.
- MAR :- It holds the address of the location that is to be accessed
- There are n general purpose registers that is R0 to Rn-1

**Performance :-**

. Performance means how quickly a program can be excecuted.
. In order to get the best performance it is required to design the compiler, machine instruction set & hardware in a coordinated manner.




## Connection B / W Processor & Memory

Connection B/W Processor & Memory

- The above mentioned block diagram consists of the following components

1) Memory
2) MAR
3) MDR
4) PC
5) IR
6) General Purpose Registers
7) Control Unit
8) ALU

- The instruction that is currently being executed is held by the Instruction Register.
- IR output is available to the control circuits, which generates the timing signal that control the various processing elements involved in executing the instruction.
- The Memory address of the next instruction to be fetched and executed is contained by the Program Counter.
- It is a specialized register.
- It keeps the record of the programs that are executed.
- Role of these registers is to handle the data available in the instructions. They store the data temporarily.
- Two registers facilitate the communication with memory.

These registers are:

1) MAR (Memory Address Register)
2) MDR (Memory Data Register)

Memory Address Register:

- The address of the location to be accessed is held by MAR.

Memory Data Register:

- It contains the data to be written into or to be read out of the addressed location.

**Working Explanation**

A **PC** is set to point to the first instruction of the program. The contents of the **PC** are transferred to the **MAR** and a Read control signal is sent to the memory. The addressed word is fetched from the location which is mentioned in the **MAR** and loaded into **MDR**. This post thus contains all the important basic operational concepts.

**EXAMPLE: Instruction cycle of Add R1 R2 R3:**

1. Instruction→ Add R1 R2 R3 i.e. R1=R2+R3

2. Add R1 R2 R3
Address → 2000

3. So PC contains the value: PC→ 2000

4. PC stores this address to the MAR so it contains the value MAR → 2000

5. Then the address location 2000 is enabled and content (Add R1 R2 R3) of Location 2000 is to MBR → Add R1 R2 R3

6. After that, it will be reached to the IR → Add R1 R2 R3

7. Decoding is performed and identifies the instruction: so opcode is 'Add' and the first, second, and third operands are R1, R2, and R3 respectively. (necessary Control signals will be generated from the control unit- identify that it is an addition operation)

8. Then R2's data will reach ALU and Then R3's data will also reach ALU, and corresponding 'ADD 'signal will be activated finally get the result to the temporary register and it is being stored back

---

## M1.03 Explain the bus structure

Bus Interconnection

- A bus is a communication pathway connecting two or more devices.
- A key characteristic of a bus is that it is a shared transmission medium.
- Multiple devices connect to the bus, and a signal transmitted by any one device is available for reception by all other devices attached to the bus.
- If two devices transmit during the same time period, their signals will overlap and become garbled.
- Thus, only one device at a time can successfully transmit.
- But at a time a single bit can be transmitted through a single bus line
- For example an 8-bit unit of data can be transmitted over eight bus lines.
- A bus that connects major computer components (processor, memory,

I/O) is called a **system bus**

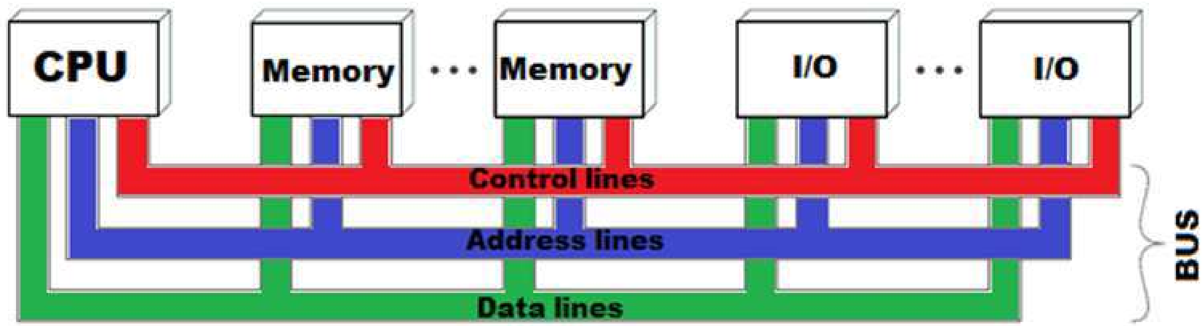
Bus Structure

• A system bus consists, typically, of from about fifty to hundreds of separate lines.

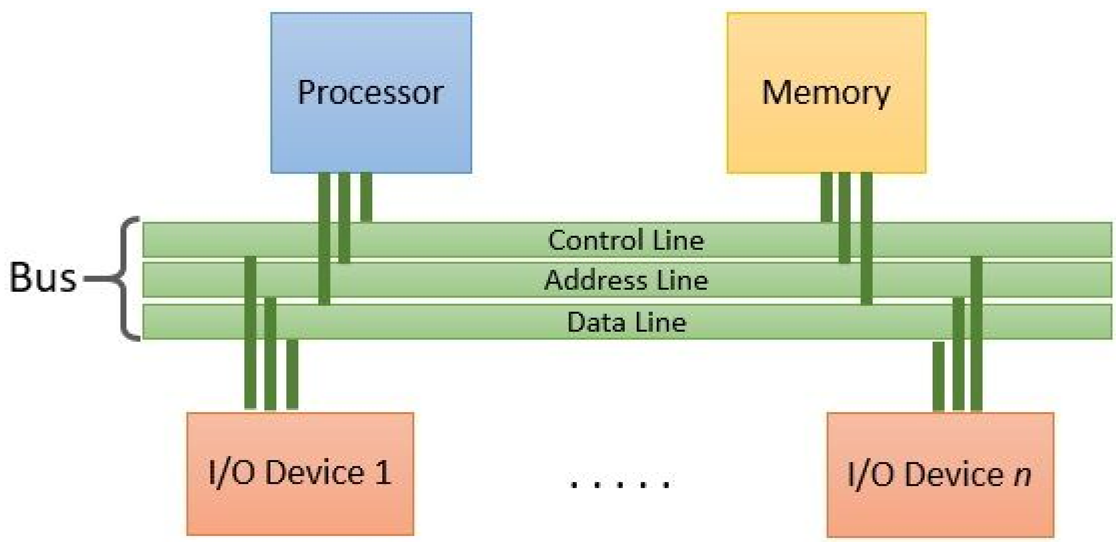
• Each line is assigned a particular meaning or function.

• The lines can be classified into three functional groups **data, address, and control lines.**

• In addition, there may be power distribution lines that supply power to the attached modules.




**Bus Interconnection Scheme**




Bus Structure

**<u>Data Bus</u>**

• The data lines provide a path for moving data among system modules. These lines, collectively, are called the **data bus**.

• The data bus may consist of 32, 64, 128, or even more separate lines, the number of lines being referred to as the width of the data bus.

• Because each line can carry only 1 bit at a time, the number of lines

determines how many bits can be transferred at a time.
• The width of the data bus is a key factor in determining overall system performance.

**Address Bus**

• The address lines are used to designate the source or destination of the data on the data bus.
• For example, if the processor wishes to read a word (8, 16, or 32 bits) of data from memory, it puts the address of the desired word on the address lines.
• Clearly, the width of the address bus determines the maximum possible memory capacity of the system

**Control Bus**

• The control lines are used to control the access to and the use of the data and address lines
• Because the data and address lines are shared by all components, there must be a means of controlling their use.
• Control signals transmit both command and timing information among system modules.
• **Timing signals** indicate the validity of data and address information.
• **Command signals** specify operations to be performed.
• If one module wishes to send data to another, it must do two things:
(1) obtain the use of the bus
(2) transfer data via the bus.
• If one module wishes to request data from another module, it must
(1) obtain the use of the bus
(2) transfer a request to the other module over the appropriate control and address lines.

It must then wait for that second module to send the data.

**Typical control lines include:**
• **Memory write:** Causes data on the bus to be written into the addressed location
• **Memory read:** Causes data from the addressed location to be placed on the bus
• **I/O write:** Causes data on the bus to be output to the addressed I/O port
• **I/O read:** Causes data from the addressed I/O port to be placed on the bus
• **Transfer ACK:** Indicates that data have been accepted from or placed on the bus
• **Bus request:** Indicates that a module needs to gain control of the bus
• **Bus grant:** Indicates that a requesting module has been granted control of the bus

- **Interrupt request:** Indicates that an interrupt is pending
- **Interrupt ACK:** Acknowledges that the pending interrupt has been recognized
- **Clock:** Is used to synchronize operations
- **Reset:** Initializes all modules.

## M1.06 Outline the memory hierarchy with respect to speed, size and cost.

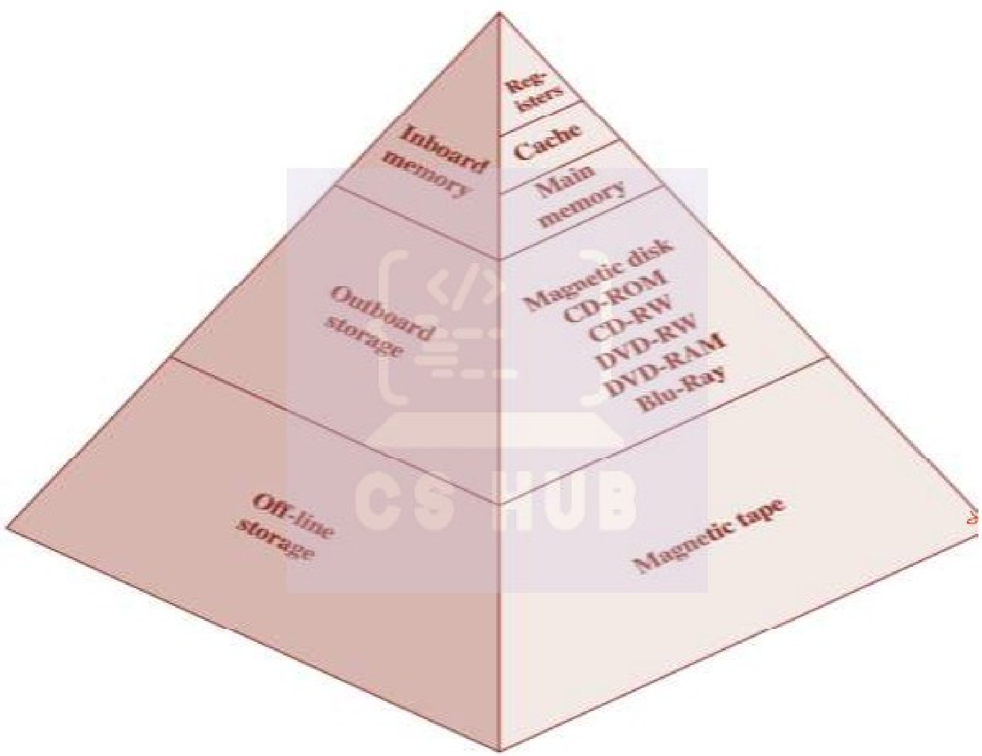
### DESIGN CONSTRAINTS ON MEMORY

- The design constraints on a computer's memory can be summed up by three questions:
- How much?
  - **Capacity**
- How fast?
  - **Access Time**
- How expensive?
  - **Cost**

- There is a trade-off among the three key characteristics of memory: capacity, access time, and cost.
- A variety of technologies are used to implement memory systems, and across this spectrum of technologies, the following relationships hold
  - Faster access time, greater cost per bit
  - Greater capacity, smaller cost per bit
  - Greater capacity, slower access time

- The dilemma facing the designer is clear.
- The designer would like to use memory technologies that provide for large-capacity memory, both because the capacity is needed and because the cost per bit is low.
- However, to meet performance requirements, the designer needs to use expensive, relatively lower-capacity memories with short access times.
- The way out of this dilemma is **not to rely on a single memory** component or technology, but to **employ a memory hierarchy**

## The Memory Hierarchy



## THE MEMORY HIERARCHY-CHARACTERISTICS

- It consists of distinct levels of memory components
- Each level characterized by its size, cost per bit and access time
- As one goes down the hierarchy, the following occur:
- Decreasing cost per bit
- Increasing capacity
- Increasing access time
- Decreasing frequency of access of the memory by the processor (Locality of Reference)
- Thus, smaller, more expensive, faster memories are supplemented by larger, cheaper, slower memories

## HIERARCHY LIST

- Registers
- L1 Cache
- L2 Cache
- Main memory
- Disk
- Optical
- Tape

- **Registers**
- The fastest, smallest, and most expensive type of memory consists of the registers internal to the processor.
- Typically, a processor will contain a few dozen such registers, although some machines contain hundreds of register

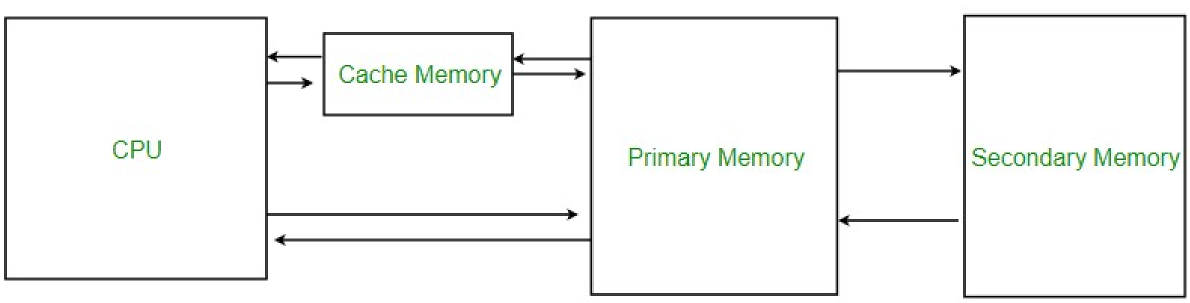
- **Cache**
- Cache is a small and fast memory situated between CPU and main memory.
- Whenever processor needs some data it first look in cache and access it .If it is not in cache it looks in to main memory.
- It is based on the principle of locality of reference
- There can be different levels like L1 cache,L2 cache
- L1 is "level-1" cache memory, usually built onto the microprocessor chip itself.
- L2 (that is, level-2) cache memory is on a separate chip(possibly on an expansion card) that can be accessed more quickly than the larger "main" memory.
- **L2 cache** is slower than **L1 cache**, but bigger in size.

- **Main Memory**
- Main memory is the principal internal memory system of the computer.
- Each location in main memory has a unique address
- Major classification include
- RAM
- ROM

- **External Memory**
- Data are stored more permanently on external mass storage devices, of which the most common are hard disk and removable media, such as removable magnetic disk, tape, and optical storage.
- External, nonvolatile memory is also referred to as secondary memory or auxiliary memory.
- These are used to store program and data files and are usually visible to the programmer only in terms of files and records, as opposed to individual bytes or words
- Different classifications
  - Disk storage
  - Optical devices
  - Magnetic tape

## M1.07 Infer the effective memory access time of cache memory

**Cache Memory** is a special very high-speed memory. It is used to speed up and synchronizing with high-speed CPU. Cache memory is costlier than main memory or disk memory but economical than CPU registers. Cache memory is an extremely fast memory type that acts as a buffer between RAM and the CPU. It holds frequently requested data and instructions so that they are immediately available to the CPU when needed.
Cache memory is used to reduce the average time to access data from the Main memory. The cache is a smaller and faster memory which stores copies of the data from frequently used main memory locations. There are various different independent caches in a CPU, which store instructions and data.



**Cache Performance:**
When the processor needs to read or write a location in main memory, it first checks for a corresponding entry in the cache.

- If the processor finds that the memory location is in the cache, a **cache hit** has occurred and data is read from cache
- If the processor **does not** find the memory location in the cache, a **cache miss** has occurred. For a cache miss, the cache allocates a new entry and copies in data from main memory, then the request is fulfilled from the contents of the cache.

The performance of cache memory is frequently measured in terms of a quantity called **Hit ratio.**

```
Hit ratio = hit / (hit + miss) =  no. of hits/total accesses
```

We can improve Cache performance using higher cache block size, higher associativity, reduce miss rate, reduce miss penalty, and reduce the time to hit in the cache.

**Application of Cache Memory –**

1. Usually, the cache memory can store a reasonable number of blocks at any given time, but this number is small compared to the total number of blocks in the main memory.
2. The correspondence between the main memory blocks and those in the cache is specified by a mapping function.

**Types of Cache –**

- **Primary Cache –**
  A primary cache is always located on the processor chip. This cache is small and its access time is comparable to that of processor registers.
- **Secondary Cache –**
  Secondary cache is placed between the primary cache and the rest of the memory. It is referred to as the level 2 (L2) cache. Often, the Level 2 cache is also housed on the processor chip.

**Locality of reference –**
Since size of cache memory is less as compared to main memory. So to check which part of main memory should be given priority and loaded in cache is decided based on locality of reference.
**Types of Locality of reference**

1. **Spatial Locality of reference**
   This says that there is a chance that element will be present in the close

proximity to the reference point and next time if again searched then more close proximity to the point of reference.

2. **Temporal Locality of reference**
   In this Least recently used algorithm will be used. Whenever there is page fault occurs within a word will not only load word in main memory but complete page fault will be loaded because spatial locality of reference rule says that if you are referring any word next word will be referred in its register that's why we load complete page table so the complete block will be loaded.